



Unit 3 · Outcome 1 · Maintaining Biodiversity

Sampling species diversity

KK06

VCE Environmental Science · Units 3 & 4 · Exam study summary

Key knowledge. 3.1.2.6 Key science skill: generate & record data Lincoln index provided in exam 3D · drag to orbit A 0.25 m² quadrat dropped on the swamp floor at Kestrel Ridge Reserve

Digital Vector EnviroSci · single-topic mastery summary with worked good-vs-bad answers, exam traps, command words and practice.



Sampling species diversity

Unit 3 · Outcome 1 · Maintaining Biodiversity · KK06 Sampling species diversity You can't count every honeyeater in the forest or every sedge on the swamp floor. So how do ecologists turn a handful of careful measurements into a believable number for the whole reserve? With grids, transects, quadrats and mark-recapture — and a healthy respect for the edges . Key knowledge 3.1.2.6 Key science skill: generate & record data Lincoln index provided in exam 3D · drag to orbit A 0.25 m² quadrat dropped on the swamp floor at Kestrel Ridge Reserve. green = counted

You measure a slice, then reason about the whole

The big idea You measure a slice, then reason about the whole A full census — counting every individual — is only possible for big, rare, obvious things (the entire wild population of Helmeted Honeyeaters is small enough to band one by one). For everything else, ecologists take a representative sample : measure a small, fairly chosen part of the habitat, then scale up. Think of tasting a pot of soup. You don't drink the whole pot to judge it — you stir it well (mix it so the sample is representative), then taste one spoonful . The trick is making sure

Quadrats — a frame of known area

Technique 1 Quadrats — a frame of known area A quadrat is a frame enclosing a known area (a classic one is 0.5 m × 0.5 m = 0.25 m²). You place it, then inside it you can record abundance, percentage cover, or which species are present. Because the area is fixed, you can convert a count into a density and scale it up to the whole habitat. Estimated total = count in quadrat quadrat area × total habitat area Below is a patch of swamp floor divided into a sampling grid. Drop quadrats and watch the estimate of total sedge plants settle down as your sample grows

Grids & the case for randomness

Technique 2 Grids & the case for randomness A grid divides the study area into equal numbered cells. You then choose which cells to sample. How you choose is the part examiners care about: Random sampling Use a random number generator to pick grid coordinates. Removes the human urge to pick "interesting" spots, so the sample is unbiased and represents the whole area. Systematic sampling Sample at fixed intervals (every 5th cell, or quadrats spaced along a line). Best when you want to capture change across a gradient rather than an overall average. The bias

Transects — sampling along a gradient

Technique 3 Transects — sampling along a gradient A transect is a line laid across a habitat. You use it when species composition changes across an environmental gradient — for example, from the wet creek edge at Kestrel Ridge, up the bank, to the dry ridge top. A random scatter of quadrats would blur that change; a transect captures it in order. Line transect Record every species that touches the line at set points. Quick; good for a first picture of zonation. Belt transect Place quadrats at intervals along the line (a strip of known width). Gives abundant

Edge effects — the boundary problem

The detail everyone forgets Edge effects — the boundary problem Real organisms straddle the frame. If you count every plant touching the quadrat, you over-count (you're stealing from the neighbouring area too). If you count none of them, you under-count. Without a rule, two students sampling the same quadrat get different numbers — the data isn't repeatable . The convention Count an individual only if it touches two chosen sides (say the top and the left). Ignore anything touching the other two sides. Over many quadrats the overcounts and undercounts can

Mark–recapture & the Lincoln index

♦ RECAP / CHEAT SHEET

Technique 4 · for animals that move Mark–recapture & the Lincoln index You can't put a quadrat around a Helmeted Honeyeater — it flies off. For mobile animals , ecologists estimate population size by capturing, marking, releasing , letting them mix, then recapturing : Capture & mark. Catch a first sample, mark them harmlessly (leg band), record the number M , release. Mix. Wait long enough for marked birds to disperse back through the population. Recapture. Catch a second sample of total size C , and count how many carry a mark, R .
$$N = M \times C / R$$
$$N = \text{estim}$$

Assumption	If it's broken...
Closed population — no births, deaths, immigration or emigration between captures	New unmarked individuals dilute the marked fraction → overestimate
Marks don't affect survival or behaviour	"Trap-shy" birds avoid recapture (R too low → overestimate); "trap-happy" birds return for the bait (R too high → underestimate)
Marks aren't lost between captures	Lost bands lower R → overestimate
Marked individuals mix randomly back in	If they don't disperse, recapture isn't representative
Enough time to mix , but not so long the population changes	Too short → clumped; too long → closed-population assumption fails

Which technique, and when?

Putting it together Which technique, and when? If you're sampling... Use... Because Plants / sessile organisms over a fairly uniform area Random quadrats on a grid They don't move; random placement gives an unbiased density & richness How a community changes across a gradient (wet→dry, near→far) Belt transect (quadrats along a line) Captures zonation in order; random quadrats would blur it A quick picture of zonation Line transect Fast record of what touches the line Mobile animals (birds, possums, fish) Mark–recapture (Lincoln index) Can't be quadratted; es

If you're sampling...	Use...	Because
Plants / sessile organisms over a fairly uniform area	Random quadrats on a grid	They don't move; random placement gives an unbiased density & richness
How a community changes across a gradient (wet→dry, near→far)	Belt transect (quadrats along a line)	Captures zonation in order; random quadrats would blur it
A quick picture of zonation	Line transect	Fast record of what touches the line
Mobile animals (birds, possums, fish)	Mark–recapture (Lincoln index)	Can't be quadratted; estimate from the marked fraction

What a full-mark answer looks like

Good vs bad · 4 marks What a full-mark answer looks like An ecologist uses mark–recapture to estimate the number of a small fish in a reserve wetland. They mark 60 fish, release them, and three days later catch 80 fish of which 15 are marked. Estimate the population and explain one reason the true number may be higher than your estimate. (4 marks) × Weak response 1 / 4 "You do 60 times 80 which is a lot, then look at the 15 marked ones. There were about 400 fish . The number could be higher because some fish died or there were more fish than they thought

Command words in this topic

◆ COMMAND WORDS

Decode the question Command words in this topic The verb tells you how much to write. Tap each one. Identify / State Describe Explain Calculate Justify / Evaluate

Sort the technique

◇ ACTIVE RECALL

Active recall Sort the technique Drag each scenario into the sampling method you'd choose. (On touch, tap a chip then tap a bin.) Nice All sorted. If any landed in red, re-read the decision rule: rooted & random → quadrats · gradient → transect · moving → mark–recapture .

Six classic traps

△ EXAM TRAPS

Don't lose easy marks Six classic traps Saying "throw the quadrat randomly". Fix: that's biased. Random means random-number coordinates on a grid . Forgetting the units. Fix: density is "per m²", estimates are "individuals" — bald numbers can drop a mark. "More accurate" when you mean "more reliable". Fix: more quadrats improve reliability/precision ; accuracy is about closeness to the true value. Confusing percentage cover with density. Fix: cover = area % (for plants you can't count); density = individuals per area. Quoting the Lincoln formula without

Practice questions

★ PRACTICE & SELF-MARK

Exam practice · self-mark · 18 marks Practice questions Write your answer, reveal the marking scheme, then award yourself marks honestly — the dial bottom-right tracks your total. These are written in VCAA style for the Maintaining Biodiversity outcome.

Build a P·E·L answer

P·E·L WRITING

Writing trainer Build a P·E·L answer Sampling questions reward a tight chain: make the Point , Explain the mechanism, then Link back to the question. Try one here. Prompt — An ecologist placed all 20 quadrats along the most flower-rich strip of meadow. Explain how this affects the validity of their density estimate and how to fix it.

P Point E Explain L Link Show a model sentence



The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

★ Good vs bad worked answers

Study the contrast — the same question answered two ways. The strong column shows the moves that earn the marks.



The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

✗ WEAK ANSWER

You do 60 times 80 which is a lot, then look at the 15 marked ones. There were about 400 fish . The number could be higher because some fish died or there were more fish than they thought. " Right ballpark by luck, but no formula shown, no working, and the "explanation" just restates the question without linking to an assumption.

✓ STRONG / MODEL ANSWER

$N = (M \times C) / R = (60 \times 80) / 15 = 320$ fish. The true number may be higher because the population is assumed to be closed , but if some marked fish died or emigrated between captures, fewer marks remain (R falls). A smaller R inflates the estimate downward relative to truth — wait, the marked fraction is too low, so the true population is larger than estimated ." Formula stated, values substituted, answer with units, and the explanation links a named assumption (closed population) to its effect on R and therefore on N.